

Auditing AlphaFold Database Comparisons: Exposure Channels and a Residue-Numbering Error

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Abstract

Comparisons between AlphaFold Database (AFDB) models and experimental structures underpin much of what is known about AlphaFold’s confidence score (pLDDT), its relation to flexibility, and the calibration of uncertainty estimates built on it. Such comparisons can go wrong in two ways that leave no visible trace. The reference structure, or a close relative, may have been available to the model during training or template search; and the residue-by-residue correspondence between model and reference may be wrong. We measure both. In the ATLAS molecular-dynamics corpus, every dated reference structure of 791 of 868 proteins (91%) was released on or before AlphaFold2’s training cutoff. In the first four batches of a preregistered census of structures released after 11 September 2026, 87 of 138 proteins (63%) already had a same-protein or near-identical structure in the training-era PDB, only 10 (7%) had no relative at 30% sequence identity, and 45 of the 138 AFDB models (33%) had been given a template of the same protein. In our own processing of the same census, a residue-numbering error, in which PDBe/SIFTS `label_seq_id` positions were applied to atoms indexed by `auth_seq_id`, changed the median IDDT of 188 of 488 chain segments (151 by more than 0.2), produced high-confidence “errors” that resemble AlphaFold overconfidence, and made a preregistered conformal-coverage prediction fail. We retract that result. On corrected labels the registered rerun meets the prediction at both levels (mean coverage 0.8945 and 0.9456 against thresholds of 0.8914 and 0.9449), narrowly at the 0.95 level; a preliminary version of this result was known before the rerun. We describe the mechanism, how often its trigger occurs, and checks that would have caught it, and we recommend reporting exposure channels separately and validating residue correspondence before any coverage claim.

1 Introduction

AlphaFold2 [Jumper et al., 2021] and the AlphaFold Protein Structure Database [Varadi et al., 2022, Bertoni et al., 2026] have made predicted structures available for hundreds of millions of proteins. A large body of work compares AFDB models with experimental structures of the same protein: to ask how well pLDDT tracks local error, whether it reflects flexibility or disorder [Akdel et al., 2022, Ruff and Pappu, 2021, Guo et al., 2022, Saldaño et al., 2022, Carugo, 2023], and, more recently, whether conformal intervals built on pLDDT reach their nominal coverage [Shihab et al., 2026]. These analyses rest on two assumptions that are rarely checked together.

The first is about exposure. For some questions, such as describing AFDB as it is used in practice, it does not matter whether the model has seen the reference. For others, such as estimating accuracy on new proteins or the calibration of pLDDT on unseen folds, it matters a great deal: if the reference or a close relative was in the training set or among the templates, agreement may reflect recall rather than generalisation, the familiar problem of leakage [Kaufman et al., 2012, Kapoor and Narayanan, 2023]. Structure prediction has a long tradition of guarding against this through blind and time-split evaluation [Kryshtafovych et al., 2021, Haas et al., 2018, Jumper et al., 2021], and recent benchmarks show that release-date splits alone still leak through homology [Buttenschoen et al., 2024, Durairaj et al., 2024, Dobson et al., 2025, Bushuiev et al., 2024]. We apply the same reasoning to retrospective AFDB comparisons, where the model is fixed and the question is what it could have seen.

The second assumption is that residues are paired correctly. Every per-residue comparison depends on a correspondence between model residues (UniProt numbering) and reference atoms (PDB numbering). PDB entries carry two residue numbering schemes, `label_seq_id` and `auth_seq_id` [Westbrook et al., 2022, RCSB Protein Data Bank, n.d.a], and mapping PDB chains to UniProt has long been a source of errors [Martin, 2005, Velankar et al., 2013]. SIFTS records its residue mappings in `label_seq_id` numbering and notes the complications added by missing residues, insertions, expression tags and linkers [Choudhary et al., 2023], and PDBrenum rennumbers PDB files to match UniProt [Faezov and Dunbrack, 2021]. The hazard itself is known; less is known about what a simple fallback between the two schemes does to a downstream analysis. A shifted correspondence does not stop a pipeline; it yields numbers that look like results.

We report an instance of each problem, measured on the same data.

1. In ATLAS [Vander Meersche et al., 2024], a molecular-dynamics corpus that has been reused to train and evaluate structure-ensemble generators [Jing et al., 2024], 791 of 868 proteins have only references released on or before AlphaFold2’s training cutoff (Section 5).
2. In the first four batches of a preregistered census of PDB entries released after a fixed date in 2026, most proteins already had a same-protein or near-identical structure in the PDB of the training era, and the AFDB model files show that 45 of 138 models were given a same-protein template (Section 6).
3. In our processing of that census, a fallback between two residue numbering schemes shifted whole chain segments, created residues with high pLDDT and low accuracy, and a preregistered coverage prediction computed on those labels failed. We describe the mechanism, its scope, and the checks that would have caught it (Section 7, Figure 1). We retract the original result; on corrected labels the prediction holds, narrowly at the 0.95 level.

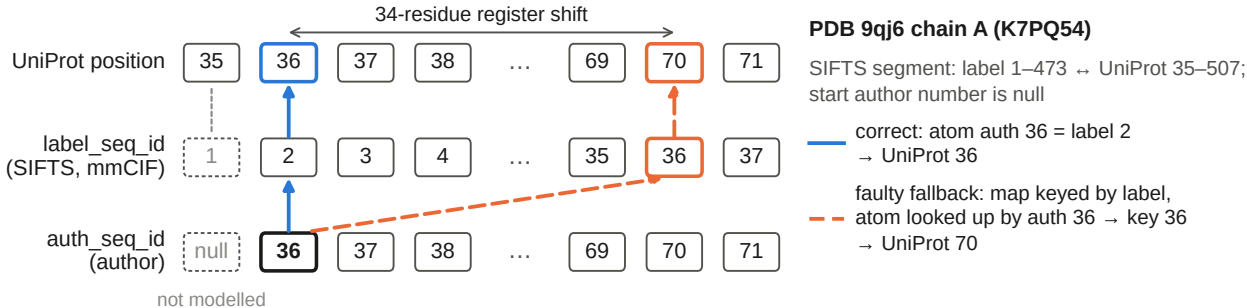


Figure 1: How the numbering fallback shifted entry 9qj6, chain A. The SIFTS segment begins at an unobserved residue, so its author number is missing and the map is built from label numbers. Atoms are then looked up by author number, which runs 34 ahead, and author residue 36 is paired with UniProt 70 instead of 36.

Throughout, the *training-era PDB* means entries released on or before 2018-04-30, and a *segment* is one contiguous stretch of a chain mapped to UniProt by SIFTS.

2 Related work

Leakage in ML for science. Leakage is a recurring cause of irreproducible results in machine-learning-based science [Kaufman et al., 2012, Kapoor and Narayanan, 2023], and biology has its own guidance on validation and splitting [Walsh et al., 2021, Whalen et al., 2022]. In structural biology, standard train–test splits were shown to leak through sequence and structural similarity in protein–ligand affinity prediction [Li and Yang, 2017, Volkov et al., 2022, Li et al., 2023, Graber et al., 2025], and date-based or metadata-based splits leave similar overlap in docking [Buttenschon et al., 2024] and protein–protein interaction benchmarks [Bushuiev et al., 2024, Kovtun et al., 2024]. Co-folding methods appear to memorise ligand poses seen in training [Škrinjar et al., 2025].

Exposure in AlphaFold evaluation. The AlphaFold2 paper evaluated on PDB chains released after the training cutoff and examined template similarity [Jumper et al., 2021]. BETA provides version-specific benchmark sets with sequence and structure filters [Dobson et al., 2025]. Comparisons of AlphaFold models with new crystal structures that control for templates [Terwilliger et al., 2024], evidence that AlphaFold reproduces memorised fold-switched states [Chakravarty et al., 2024], and training-set ablations [Ahdritz et al., 2024] together indicate that what the model has seen changes what it predicts.

Conformal prediction for structure. Split conformal prediction gives finite-sample marginal coverage under exchangeability [Vovk et al., 2005, Lei et al., 2018, Angelopoulos and Bates, 2023]. Residues of the same protein are not exchangeable with residues of other proteins, and hierarchical data need their own guarantees [Dunn et al., 2023, Lee et al., 2023, Barber et al., 2023]. CalPro applies evidential and conformal methods to pLDDT [Shihab et al., 2026]; we do not reproduce its results here.

3 Exposure channels

Let R be an experimental reference structure, first released on date r , compared with an AFDB model M of the same UniProt accession. All models in the preregistered census are AlphaFold2 monomer models (the model files identify them as “AlphaFold Monomer v2.0”); a few ATLAS accessions are served by ColabFold models. We distinguish three ways in which M could have been informed by R or by structures like it.

Training. AlphaFold2 was trained on PDB structures released on or before 2018-04-30 [Jumper et al., 2021]. If r is on or before that date, R may have been a training example. Training also filtered and sampled chains by cluster and used self-distillation, so eligibility is an upper bound on use.

Templates. At inference time AlphaFold2 searches the PDB for templates. The human-proteome study reports the PDB and PDB70 snapshots it used [Tunyasuvunakool et al., 2021], and AFDB states that templates were restricted to structures released before 2021-02-15 [EMBL-EBI AlphaFold Database, n.d.]. AFDB model files list the PDB entries supplied as templates (`_ma_template_ref_db_details`), so this channel can be measured per model. The list records which templates the network was given, not how much the prediction relied on them.

Homology. Even when R post-dates both cutoffs, the same protein or a close homologue may have been in the PDB before the training cutoff. Release dates do not measure this channel; sequence similarity to the pre-cutoff PDB does [Jumper et al., 2021, Buttenschoen et al., 2024]. Sequence information also reaches the model through multiple sequence alignments and self-distillation; we do not measure that.

AFDB entries also carry a `modelCreatedDate`. AFDB documents it as the date the entry was created [Google DeepMind, n.d.], and in version 6 the unchanged models received new metadata and the v6 label while their coordinates were carried over from version 4 [PDBe, 2025]. It is not a record of when a prediction ran or what it could see, so we do not use it as an exposure measure; the census below uses it only to exclude models created after its start date.

4 Data and methods

ATLAS. The ATLAS corpus [Vander Meersche et al., 2024] contains 1,938 entry-chain records over 1,735 PDB entries and 943 UniProt accessions, of which 869 have both an AFDB model and a SIFTS mapping [Velankar et al., 2013, Dana et al., 2019]. For each reference we took the initial release date (the first entry of the `mmCIF_pdbx_audit_revision_history`). One accession has an AFDB record we could not decode, leaving 868 accessions and 935 accession-entry pairs.

Preregistered census. Before any label existed, we froze a census design, its analysis code and its predictions in the project repository and timestamped them with OpenTimestamps. The design fixes an AFDB version-6 snapshot at $T_0 = 2026-09-11\ 00:00$ UTC and collects, in weekly batches, X-ray entries at 2.5 Å

resolution or better released after T_0 . Models created on or after T_0 are excluded. This design does not add exposure control beyond what any post-2021 release filter gives an AlphaFold2 model; its value is that every choice in the design was fixed before any IDDT value was computed. The first four weekly batches (the first was empty), ending 4 October 2026, produced 488 chain segments from 272 accession–entry pairs and 138 accessions under the original residue mapping (495 segments and 141 accessions after the correction in Section 7). We report these four of the 12 planned weekly batches. The unit of analysis for exposure is the accession; for labels it is the residue, grouped by chain segment.

Post-2022 census. An earlier exploratory census of 168 X-ray entries released after 2022-06-02 used the same pairing code, except that it discarded pairs whose amino acids differed. We use it to check whether findings from the preregistered census recur.

Residue pairing and IDDT. We call the per-residue IDDT values *labels*. For each reference chain we pair C α atoms with AF model residues through the PDBe SIFTS UniProt mapping. The corrected labeller keys the mapping on the label residue number (`label_seq_id`) and the chain’s label identifier (`struct_asym_id`); maps equal-length segments by offset and unequal-length segments by global alignment of the entity sequence to the UniProt sequence; drops any pair whose amino acids differ; and rejects a segment if more than 10% of its pairs differ. Chains with fewer than 30 aligned residues are excluded. The label is IDDT-C α [Mariani et al., 2013], computed per chain against the AF monomer over aligned residues, with a 15 Å neighbour radius and thresholds of 0.5, 1, 2 and 4 Å. pLDDT is read from the model’s B-factor column and never used to build labels.

Homology exposure. For each census accession we queried the RCSB search service [Burley et al., 2023] for entries released on or before 2018-04-30 that (i) map to the same UniProt accession, (ii) contain a polymer at 95% or higher sequence identity, or (iii) at 30% or higher (MMseqs2 [Steinegger and Söding, 2017]; e-value ≤ 0.1 ; query = canonical UniProt sequence; no query-coverage threshold, so a hit, including a same-accession hit, may cover only part of the protein). Each accession takes its strongest class. A failed query leaves an accession unresolved rather than novel; none failed (Appendix B).

Template exposure. From each AFDB model file the census downloaded, we read the template PDB IDs, mapped each to UniProt through SIFTS, and recorded its release date. A model has a same-protein template if any template maps to its accession.

Registered coverage prediction. For each of 200 random protein-level splits (seeds 0–199), 50% of accessions form a training set used only to compute the median IDDT $\hat{y}$, 25% form the calibration set, and 25% the test set. Every residue receives the interval $\hat{y} \pm q$, where q is the $\lceil (n_{\text{cal}} + 1)(1 - \alpha) \rceil$ -th smallest calibration score $|y - \hat{y}|$ [Vovk et al., 2005, Lei et al., 2018]. The interval ignores pLDDT, so the test concerns the pipeline, not AlphaFold’s confidence. We predicted that mean residue-level test coverage over the 200 splits would be at least the nominal level minus three Monte Carlo standard errors, at $1 - \alpha = 0.90$ and 0.95; the MCSE is the standard deviation of coverage over splits divided by $\sqrt{200}$, so the threshold depends on the data. Calibration residues are pooled across proteins of very different sizes, and for such hierarchical data marginal residue-level coverage carries no finite-sample guarantee [Dunn et al., 2023, Lee et al., 2023].

5 Most ATLAS references predate the training cutoff

Table 1: Training eligibility of ATLAS references. An accession counts as training-eligible when every dated reference was released on or before 2018-04-30.

Unit	Total	Training-eligible	At least one later reference
Accessions	868	791 (91.1%)	77
Accession–entry pairs	935	851 (91.0%)	—

Of 868 ATLAS proteins, 791 (91.1%) are represented only by structures released on or before AlphaFold2’s training cutoff (Table 1). The 935 dated references were released between 1988 and 2023, with a median of 2008. Agreement between AFDB models and these references cannot by itself be read as evidence of generalisation; only 77 proteins have any reference released after the cutoff. Comparing release dates with the AFDB creation date instead would flag 863 of 868 proteins (99.4%), a figure that reflects when database entries were created rather than what the model saw. ATLAS was built to describe dynamics, not to evaluate AlphaFold; reusing it to study AlphaFold error needs an explicit exposure control.

6 A post-cutoff release does not imply an unseen protein

Table 2: Exposure of the 138 proteins in the first four batches of the preregistered census. Homology: strongest relative in the PDB released on or before 2018-04-30. Templates: what the AFDB model file lists.

Class	Proteins	Share
<i>Homology (training-era PDB)</i>		
Same UniProt accession	78	56.5%
Other entity, $\geq 95\%$ identity	9	6.5%
Other entity, $\geq 30\%$ identity	41	29.7%
No relative at $\geq 30\%$	10	7.2%
<i>Templates (AFDB model file)</i>		
Same-protein template	45	32.6%
Other templates only	88	63.8%
Unresolved ^a	5	3.6%

^a The SIFTS lookup failed for at least one template, and no other template maps to the protein.

Every structure in the preregistered census was released after T_0 , more than eight years after the training cutoff. Yet 87 of 138 proteins (63.0%) had a structure of the same accession or of a 95%-identical sequence in the training-era PDB, 128 (92.8%) had a relative at 30% identity or more, and only 10 (7.2%) had none (Table 2, Figure 2). Well-studied proteins have many: hen egg-white lysozyme (P00698) had 730 same-accession entries before the cutoff, human carbonic anhydrase 2 (P00918) 700, and β_2 -microglobulin (P61769) 693. These sequence hits show that a related structure existed; they do not say whether it differs by ligand, mutation, construct or crystal form.

The template records give a more direct measure. All 138 model files list four PDB templates, released between 1988 and January 2021, none after the documented 2021-02-15 limit. For 45 proteins (32.6%) at least one template is a structure of the same protein; for 88 all templates are other proteins; for 5 the SIFTS lookup failed for at least one template and no other template maps to the protein. The two channels overlap but do not coincide. Of the 78 proteins with a same-accession structure in the training-era PDB, 43 were given a same-protein template and 35 were not. Conversely, two proteins with no same-accession structure before the training cutoff were given one as a template, released in 2019 and 2020; one of them (Q9F0J8) has no relative at 30% identity in the training-era PDB at all.

Does exposure change agreement? Prevalence alone does not show that exposure matters. As an exploratory check, registered before it was run, we compared per-protein agreement in the post-2022 census between proteins with a close training-era relative (same accession or $\geq 95\%$ identity) and those without, using each protein’s median IDDT and median pLDDT and averaging over proteins. The 40 proteins with a close relative had a slightly higher median IDDT than the 45 without (0.975 against 0.959; difference 0.016, 95% bootstrap interval 0.003 to 0.032), but also a higher median pLDDT (96.1 against 92.3; difference 3.8, interval 1.7 to 6.0). The per-protein share of confident errors (pLDDT ≥ 90 and IDDT < 0.60) was the same in both groups (0.12% and 0.13%; difference -0.01 percentage points, interval -0.18 to 0.17). At this sample size, proteins with close relatives are predicted somewhat better and pLDDT reflects this; we see no sign that exposure distorts pLDDT calibration, but the comparison could not detect a small effect. We did

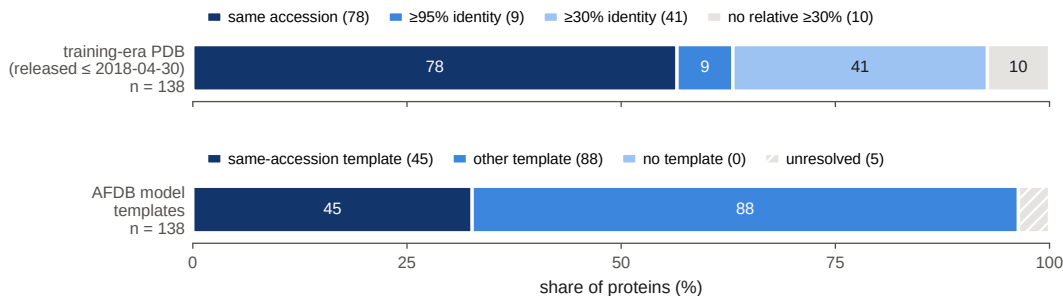


Figure 2: Exposure of the 138 preregistered-census proteins: strongest relative in the training-era PDB (top) and templates listed in the AFDB model file (bottom).

not run this comparison on the preregistered census, where pLDDT-conditional coverage is the subject of registered predictions that have not yet been evaluated.

7 A residue-numbering error that manufactured confident errors

7.1 Mechanism

PDBe’s SIFTS mapping gives, for each segment of a chain, its start and end in `label_seq_id` numbering and, when that residue was observed, in author (`auth_seq_id`) numbering. Our census code built each residue map from the author number and fell back to the label number when the author number was missing, which happens whenever the first residue of a segment is unobserved. Atoms were then looked up by author number. When the two schemes differ by an offset, every residue of the segment is paired with the wrong UniProt position. No per-residue identity check was in place to catch it.

Figure 1 shows entry 9qj6, chain A, of K7PQ54 (dihydrolipoyl dehydrogenase from the lesser grain borer, 507 residues). The SIFTS segment starts at label residue 1, UniProt 35, which is unobserved. Author numbering runs 34 ahead of label numbering, so the atom with author number 36 (label 2, UniProt 36) was assigned UniProt 70. Under this shift the chain’s median IDDT was 0.198 (chain B: 0.199) at a median pLDDT of 98.4. With the mapping fixed, 472 and 471 residues align with no sequence mismatch and median IDDT 0.993 and 0.995.

7.2 Scope

We replayed the census from its stored, hash-checked source files with the original and the corrected labeller. With the original labeller the replay reproduced every stored batch exactly, which confirms that it re-executes the registered pipeline. Table 3 compares the two label sets.

Table 3: The first four batches of the preregistered census, labelled with the original and the corrected residue mapping.

	Original	Corrected
Residue rows	117,905	123,454
Chain segments	488	495
Accessions	138	141
Residues with IDDT < 0.60	31.9%	1.1%
High-confidence, low-accuracy chains ^a	116	0
K7PQ54 chains with median IDDT < 0.4	29 of 38	0 of 38

^a At least 30 residues, median pLDDT ≥ 90 and median IDDT < 0.4; the 116 chains span 32 accessions.

Of the 488 original chain segments, 188 changed in median IDDT and 151, across 40 accessions, changed by more than 0.2 (Figure 3). The share of residues below 0.60 IDDT fell from 31.9% to 1.1%. After correction

no chain combines high pLDDT with low IDDT, and none of the 13 entries whose chains had disagreed with each other by 0.4 or more in median IDDT still does. K7PQ54 alone contributed 14% of the original rows and half of the misses at the 0.90 level. The corrected set is not a subset of the original: it labels 6,465 residue positions that the original did not and drops 916 that it did (a net gain of 5,549), and it adds three accessions and seven segments that the original mapping had rejected. An independent audit with a separately written parser and IDDT implementation reproduced the corrected labels of 36 sampled segments exactly.

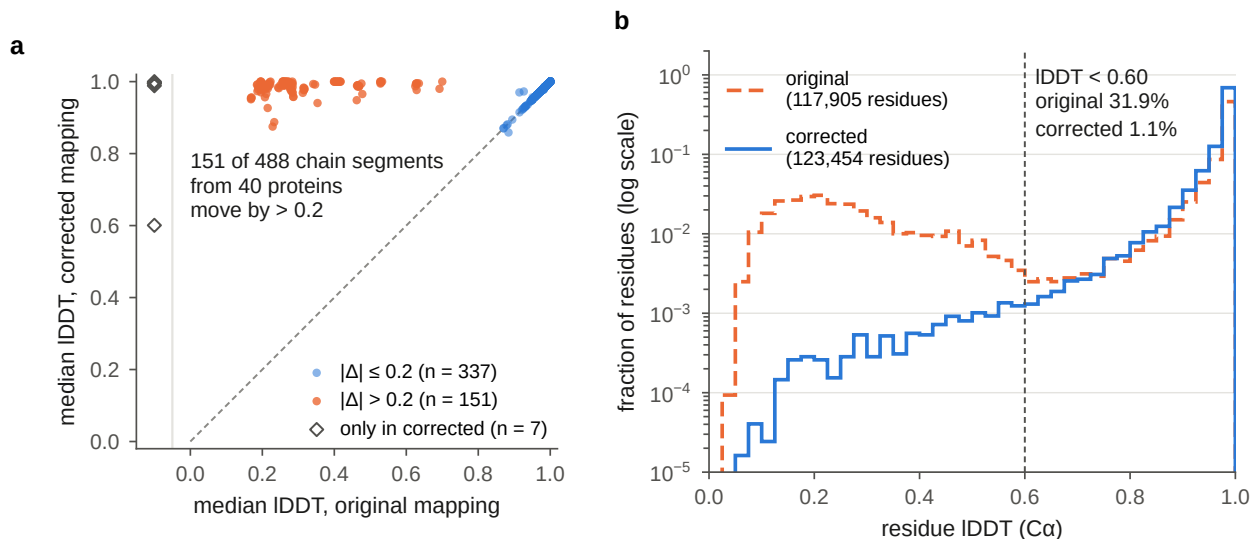


Figure 3: Effect of the correction on the first four batches. (a) Median IDDT per chain segment with the original (x) and corrected (y) mapping; the dashed line marks no change, and segments that changed by more than 0.2 are shown in orange. Diamonds at the left are the seven segments that only the corrected mapping labels. (b) Distribution of residue-level IDDT with each mapping (logarithmic scale).

7.3 Why it was dangerous, and how it could have been caught

The error did not add noise. It created residues that AlphaFold was confident about and that appeared to be badly wrong, which is the signature of overconfidence. A calibration study run on these labels could have reported that AlphaFold is overconfident on new structures, and our registered coverage test did fail on these labels. The registration and the stored source files are what let us trace the failure to the labels rather than to the model.

The precondition is common: 304 of the 492 SIFTS segments mapped to these proteins (62%) begin on an unobserved residue. It becomes an error only when author and label numbering are offset, which was the case for 147 segments (30%) from 39 proteins, with offsets from -31 to $+201$. In a comparison we added after the fact, all 147 were among the 151 segments whose median IDDT changed by more than 0.2; in the other four the offset changes within the segment.

The direct test is an amino-acid identity check on every pair: under a register shift almost no paired residues match, which is why the post-2022 census, which discarded mismatched pairs, lost data instead of corrupting it (Section 7.4). Coarser checks catch some cases. A third of residues below 0.60 IDDT is far from the 2.6% seen in the corrected post-2022 census, and 13 entries had chains of the same protein that disagreed by 0.4 or more in median IDDT. A chain-agreement check would still miss entries such as 9qj6, where both chains were shifted alike.

7.4 The same error in the post-2022 census

Because the post-2022 census discarded pairs whose amino acids differed, the error there mostly removed data rather than corrupting it: of 168 entries, 69 were rejected for having too few aligned residues and 3

because the model could not be downloaded. With the corrected mapping, 45 of those 69 produce labels, and 139 entries in all instead of 96 (37,531 residue rows instead of 24,457). The rank correlation between pLDDT and lDDT rose from 0.617 to 0.643. Among residues with $\text{pLDDT} \geq 90$, the share below 0.60 lDDT halved, from 0.58% ($n = 20,777$) to 0.29% ($n = 31,032$), while in the lowest-confidence bin ($\text{pLDDT} < 50$) it rose from 58% ($n = 177$) to 75% ($n = 314$). Most of these changes come from the 45 newly labelled entries; residues that matched by chance under a shift, and so passed the identity filter, account for a smaller part (for example in entry 10ke). This relabel is exploratory. It used source files fetched again in October 2026, because the September files had not been kept, whereas the preregistered census was replayed from its original stored files.

7.5 The registered coverage prediction, before and after

Table 4: Registered coverage prediction on the original and corrected labels: mean residue-level test coverage over 200 protein-level splits and the threshold, nominal minus three Monte Carlo standard errors. The prediction fails on the original labels and holds on the corrected labels.

Nominal	Original labels		Corrected labels	
	Mean coverage	Threshold	Mean coverage	Threshold
0.90	0.8580	0.8760	0.8945	0.8914
0.95	0.9152	0.9328	0.9456	0.9449

On the original labels the prediction failed at both levels (Table 4). Those labels were corrupted, and we retract the result; the original outputs remain archived unchanged. On the corrected labels (123,454 residues from 141 accessions, against 117,905 from 138 before) the prediction holds at both levels. At 0.90 the mean exceeds the threshold by 0.0031, about one Monte Carlo standard error; at 0.95 it does so by 0.0007, less than half of one (MCSE 0.0017), so this level passes only narrowly. A margin this small could reverse under a different but equally defensible repair. It does not reverse when residues on the interval boundary are counted as uncovered (0.9453 against 0.9449).

The rerun is not blind. Before registering it we computed the same statistic outside the registered pipeline on a preliminary version of the repair (122,024 residues rather than 123,454) and obtained 0.898 and 0.947. The seeds and decision rule were fixed by the original registration (the thresholds depend on the data only through the MCSE), and the repair rule was registered before the rerun. The rerun used the registered evaluator, called directly because data collection was paused while the error was fixed. An independent reimplementaion that shares no code with the evaluator reproduced both means exactly and, on the original labels, the original failure. The original failure is therefore consistent with a labelling artefact. Because the interval ignores pLDDT, the result is consistent with sound labels and a sound pipeline; it says nothing about how well pLDDT tracks error.

8 Recommendations

1. Report training eligibility, template exposure and homology exposure separately, and state which ones the question requires to be controlled. Release dates and database creation dates are not exposure measures.
2. For AFDB models, read the template list from the model file; it is available and cheap to extract.
3. Stratify results by sequence similarity to the training-era PDB, as leakage-aware benchmarks already do.
4. Key residue correspondences on a single numbering scheme, bind them to the chain identifier, and assert amino-acid identity for every pair.
5. Sanity-check label distributions against pLDDT and between chains of the same entry before running any downstream analysis.

6. Register analyses before labels exist and keep the raw inputs, so that an error found later can be replayed and corrected rather than argued about.

9 Limitations

First, the census models are AlphaFold2 monomer models; other versions and AlphaFold3 [Abramson et al., 2024] have their own cutoffs. Second, template lists record what the network was given, not what it used, and we did not measure sequence-level exposure through alignments or self-distillation. Third, homology was measured with fixed identity thresholds, a single canonical sequence and no coverage threshold, so a 30% hit can be a short domain match. Fourth, labels compare single chains with monomer models, so interfaces and ligands in the crystal are ignored. Fifth, the preregistered census is reported after four of its 12 weekly batches (138 proteins, 10 without a relative), and the full cohort may differ. The corrected label set also differs from the original (6,465 residue positions gained, 916 lost). Finally, the post-2022 relabel was run on re-fetched source files, which may differ from those used originally.

10 Conclusion

In both corpora we examined, most proteins were exposed in at least one channel. Exposure has to be measured channel by channel; the template lists in AFDB model files make one of those channels directly observable. Residue correspondence deserves the same care. In our case a single numbering fallback was enough to make accurate predictions look like confident errors and to make a registered coverage prediction fail; on corrected labels the same prediction holds, narrowly at the 0.95 level. The registration and the stored inputs let us find the cause and undo it.

A Provenance

Table 5: Timeline of the preregistered census and its correction.

Date (2026)	Event
9 Sep	Design, analysis code and predictions frozen (Bitcoin timestamp confirmed 10 Sep)
11 Sep	Snapshot instant T_0
4–5 Oct	Batch-4 evaluation records the coverage failure
7 Oct	Independent audit traces 9qj6 to the numbering fallback; collection paused
7 Oct	Repair, replay protocol and correction registered before rerunning
7 Oct	Replay of batches 1–4 with both labellers; post-2022 census relabelled
7 Oct	Independent audit of repair, replay and relabel
9 Oct	Registered rerun on corrected labels completes; independent recomputation

Internally the analyses are numbered e420 (ATLAS), e421 (post-2022 census), e422 (preregistered census) and e427 (repair and replay). Registrations and checkpoints carry SHA-256 digests timestamped with OpenTimestamps.

B Homology search details

Queries used the RCSB Search API [RCSB Protein Data Bank, n.d.b]. Every request and response was cached with its SHA-256 digest. The API returns HTTP 204 for a query that ran and found nothing; across the 224 accessions queried for both censuses, 594 queries returned 200, 408 returned 204, and none failed.

Data and code availability

The registrations with their timestamp proofs, the analysis code, the derived residue-level tables and the request digests are available at <https://github.com/MachineLearning-Nerd/afdb-exposure-audit>, together with a script that recomputes every number in this paper from them. The full evaluation records of the preregistered census, which contain predictions that are still blind, will be added when the census closes. Raw files from the PDB, AFDB, PDBe/SIFTS, UniProt and ATLAS are not redistributed; each can be re-fetched from its public source, and the SHA-256 digests recorded with every request identify the exact bytes used. The preregistered census was replayed from its original stored files; the post-2022 relabel used files fetched again in October 2026 (Section 7.4).

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